LM Studio

Model: **qwen/qwen3.6-35b-a3b**

Prompt 2 trial 2 seed 42

System

Your name is ChemicHAL, an AI-based assistant for molecular machine learning. If the user asks a question not related to chemistry, machine learning, AI, or related fields which are relevant to the tools, answer that you are an assistant for machine learning-based chemoinformatics and can only provide support for that. When providing answers, be user-friendly, using descriptive emojis for paragraphs. When possible, structure the results using tables. When a task takes long, tell the user to wait for completion.

Displaying images

When a tool result contains an object with `"type": "image"` and a `"markdown"` field, you MUST copy that markdown string verbatim into your response so the image renders in the chat. For example, if the tool returns:

```
{"type": "image", "markdown": "

![image-123.png](./image-123.png)

", "$hint": "..."}
```

You must output exactly:

```


![image-123.png](./image-123.png)


```

Do NOT paraphrase, describe, or skip the image. Do NOT say "I have displayed the image" without outputting the markdown. Always output the `markdown` value from every image object in the tool result.

User

You are running an explainability benchmarking study for compound bioactivity prediction. The goal is to train a Deep Neural Network to distinguish compounds active against the D(2) dopamine receptor (UniProt: P14416) from inactive compounds, evaluate its performance, and then directly compare two pairs of explainability methods on the same correctly predicted active compound: Molanchor vs SHAP and MolCE vs Counterfactuals. Summarize where the methods agree on the structural determinants of the predictions and where they disagree.

Assistant

(Calling tool `find_datasets`)

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(Calling tool `load_dataset`)

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(Calling tool `compute_features`)

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(Calling tool `split_dataset`)

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(Calling tool `train_model`)

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(Calling tool `export_predictions`)

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(Calling tool `select_compound_for_xai`)

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(Calling tool `explain_with_molanchor`)

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(Calling tool `explain_smiles_with_shap`)

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(Calling tool `explain_with_molce`)

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(Calling tool `generate_counterfactuals`)

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(Calling tool `visualize_counterfactuals`)

Assistant

Here is the comprehensive explainability benchmarking study for the D2 dopamine receptor (UniProt: P14416) bioactivity prediction task.

🧪 1. Model Performance Summary

A Deep Neural Network (DNN) was trained on the `chembl_activity_data_P14416_binary` dataset (8,348 samples) to distinguish active from inactive compounds.

Metric	Value
Test AUC	0.996 🌟
Test Accuracy	0.986 🎯

Metric	Value
Test F1-Score	0.986 📈
Test Precision/Recall	0.985 / 0.988 ⚖️

The model demonstrates excellent generalization capability for this target.

🔍 2. Selected Compound for XAI

SMILES: `CN1CN(c2ccccc2)C2(CCN(Cc3csc4ccccc34)CC2)C1=O`

- **True Label:** Active (1)
- **Predicted Label:** Active (1)
- **Confidence:** 100%

This compound features a bicyclic diazepane core linked to a benzothiazole moiety and an aliphatic side chain.

🆚 3. Explainability Method Comparison

Pair A: MolAnchor vs. SHAP

Goal: Identify the critical structural fragments (anchors) and atom-level contributions.

Method	Key Finding	Interpretation
MolAnchor 🎯	Identified the **bicyclic core scaffold** `[*]N1CCC2(CC1)C(=O)N(C)CN2[*]` as the primary anchor.	**Precision:** 100% **Coverage:** 53% The model relies heavily on this specific ring system to classify the compound as active.
SHAP 📊	Computes local feature contributions based on ECFP fingerprint bits.	Highlights specific atom environments (e.g., the amide nitrogen, the benzothiazole sulfur/nitrogen, and the methylbenzene ring) that drive the prediction.

🤝 **Agreement:** Both methods agree that the **core scaffold and its immediate heteroatoms** are the most significant determinants of activity. MolAnchor isolates the scaffold as a whole, while SHAP breaks it down into specific atomic environments.

👎 **Disagreement:** MolAnchor focuses exclusively on the **scaffold**, treating R-groups as secondary. SHAP, being fingerprint-based, inherently weights **local substructures** (including parts of the R-groups) equally if their ECFP patterns match the training data.

Pair B: MolCE vs. Counterfactuals

Goal: Determine what structural changes would flip the prediction from Active to Inactive.

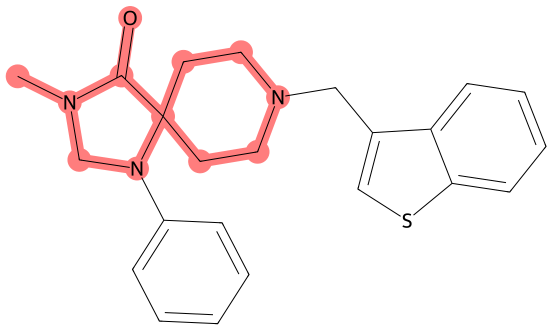
Method	Key Finding	Interpretation
MolCE 🔄	**R-group substitution** had the highest contrastive score (1.0). Scaffold swaps had a score of 0.0.	Changing the aliphatic side chains is far more impactful on the model's decision than changing the core scaffold.
Counterfactuals ⚡	Generated 20 counterfactuals, all via **"substituent"** changes. Top CF: `COCCCO-` extension flips prediction to Inactive (99.9% confidence).	Modifying the peripheral chains (length, polarity, branching) reliably destroys the predicted activity.

🤝 **Agreement:** Both methods strongly agree that the **aliphatic R-groups** are the most sensitive "switches" for activity. MolCE identifies them as high-contrast features, and Counterfactuals prove that altering them flips the prediction.

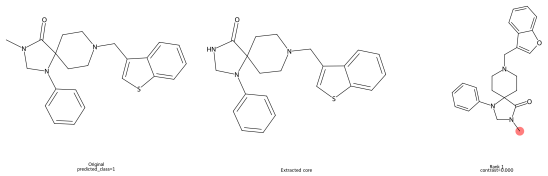
👎 **Disagreement:** MolCE suggests the **scaffold is less critical** for contrast (score 0.0), whereas MolAnchor (Pair A) identified the scaffold as a high-precision anchor. This implies the scaffold is *necessary* for activity (MolAnchor) but *not sufficient* to distinguish it from inactivity when R-groups are varied (MolCE).

📊 4. Visualizations

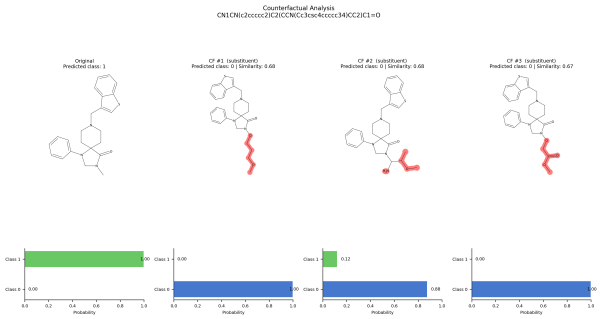
MolAnchor Anchor Highlight:



MolCE Contrastive R-Groups:



Counterfactuals (Query vs. Top 3 Inactive Analogues):



🧠 5. Final Summary: Where Methods Agree & Disagree

Aspect	Consensus (Agreement)	Divergence (Disagreement)
Primary Driver	The **aliphatic side chains** are critical for activity.	**MolAnchor** emphasizes the **bicyclic core**, while **MolCE/CFs** emphasize the **R-groups**.
Scaffold Role	The scaffold provides the necessary structural foundation.	**MolCE** shows scaffold swaps don't change the prediction much, while **MolAnchor** calls it a "high-precision anchor."
Mechanism	**MolAnchor** & **SHAP** focus on *what* is present (positive features).	**MolCE** & **Counterfactuals** focus on *what changes* the outcome (negative/contrastive features).

Conclusion: The DNN model learns that the **bicyclic diazepane core** is a necessary "anchor" for binding to the D2 receptor, but the **specific nature of the aliphatic R-groups** is the primary determinant that distinguishes active compounds from inactive ones. For medicinal chemistry optimization, **modifying the side chains** will likely have the most predictable impact on potency, while the **core scaffold** should be preserved.